

Explainable classification of seizures and other patterns of harmful brain activity in critically ill patients

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Abstract

Accurate detection and classification of seizures from electroencephalography (EEG) data can potentially enable timely interventions and treatments for neurological diseases. Currently, EEG recordings are exclusively reviewed by human experts, namely neurologists with specialized training. While indispensable, this time-consuming workflow represents a major bottleneck. Review of EEG records is laborious, time-consuming, expensive, prone to fatigue-induced errors, and suffers from inter-rater reliability even among expert reviewers. This paper introduces a new deep neural network (DNN) with interpretable layers for the classification of seizures and other pathologic brain activities such as periodic discharges, rhythmic delta waves and miscellaneous activities. The DNN architecture uses interpretable layers that allow clinicians to evaluate the model's decision-making pipeline and build trust in the model and support clinical decision making. The combination of deep learning and interpretability layers is novel and addresses the limitations of existing methods. We demonstrate the usefulness of the proposed approach on a publicly available EEG dataset. Our method achieves state-of-the-art performance and provides classification decisions that are interpretable, useful for clinical experts. This paper contributes to the existing body of literature on EEG-based seizure detection and addresses the gap between DNN-based methods and clinical interpretability, leading to accurate and clinically meaningful predictions.

Introduction

Precise classification of seizures and other patterns of abnormal brain activity in the critically ill is crucial for diagnosis, treatment, and management [1]. A large number of seriously ill patients including those in the intensive care units (ICUs) are at risk of diverse neurological ailments that can adversely influence their outcome and quality of life [2]. Seizures are serious threats to the neurological well-being and are associated with long term cognitive deficits if missed or mistaken [3]. Other patterns of abnormal brain activity including periodic discharges and rhythmic activities are associated with underlying neurological injuries and must be studied for interpretation [4]. Classification of seizures and other abnormal brain activities in the critically ill patients is challenging and multi-factorial process that considers integrated analysis of clinical symptoms, EEG findings and neuroimaging studies [5]. Numerous types of seizures including focal seizures, generalized seizures and non-convulsive seizures should be distinguished for appropriate treatment procedure and optimizing the outcome [6]. Over the past decade, there has been considerable development in the field of neurocritical care and continuous EEG monitoring has become quite widespread in the ICUs for the analysis and management of seizures and other patterns of uneven brain activity in the critically ill patients [7]. Technological developments including continuous EEG monitoring and mechanized seizure detection algorithms have made it promising to monitor brain electrical activity in real time, sense seizure activity as it happens and intervene accordingly [8]. Yet, there remains significant scope to improve the accurate analysis of EEG patterns, differentiation of pathological findings from artifacts, and optimization of seizure detection algorithms for various patient populations [9]. By advancing the recognizing of these neurologic conditions, clinicians can increase the precision of diagnosis, optimize patient treatments, and advance the neurologic outcome of critically ill

patients [10]. Extremely ill people often have seizures, which are caused by abnormal brain electrical discharges. If these discharges are not properly attended, they can lead to increased brain damage and a worse outcome [11]. Other types of dangerous brain activity, such as rhythmic and periodic activities, may also be a indicator of a deeper neurological damage and need to be properly examined and interpreted [12]. The detection and classification of these patterns should be rapid to assistance in treatment decisions and improve results for this population. Electroencephalographic (EEG) monitoring, neuroimaging studies, and clinical evaluation are all relied upon to determine whether or not seriously ill patients are undergoing seizures or other dangerous patterns of brain activity [13]. Continuous EEG monitoring has appeared as a helpful method to identify seizure activity and explain EEG patterns in real time [14]. It is significantly important to distinguish among various seizure types, including focal seizures, generalized seizures, and non-convulsive seizures, in order to guide optimum treatment decisions and improve patient outcomes [15]. The ideal management of seizures and other patterns of dangerous brain activity in seriously ill patients requires a multidisciplinary approach from neurologists, intensivists, and neurocritical care specialists [16]. Common treatment options include the use of anti-epileptic medication, such as benzodiazepines, phenytoin, or levetiracetam, to restrain seizure activity and reduce the risk of future seizures [17]. Less imposing treatments with continuous mixture of anaesthetic agents or surgical resection may be necessary in cases where other treatments have failed. Moreover, enhancing the management of the underlying medical condition, such as metabolic disruptions or intracranial hypertension, is necessary to improve neurological outcome in seriously ill patients [18]. Seizures and other patterns of dangerous brain activity are common occurrences in critically ill patients and require timely recognition and treatment to prevent harmful outcomes [19]. Continuous EEG monitoring plays an fundamental role in the diagnosis and classification of patients, allowing physicians to react appropriately and deliver optimum care [20]. Further research is needed to improve our identification of seizure categorization and develop more efficient treatment options for patients with neurological disorders who suffer acute deterioration [21]. By enhancing the knowledge in this field, clinicians may be able to enhance patient outcomes and raise the quality of care for people in severe illness. This paper is proposed to state the classification of seizures and other patterns of dangerous brain activity in seriously ill patients as well as the clinical significance, diagnostic workup, and clinical implications of these patterns. Also, it is projected to assist clinicians to classify the seizures and other patterns of dangerous brain activity in critically ill patients under several conditions.

Proposed Approach

As for the suggested model of categorization, it can be represented in Fig. 1. Here, the pipeline relates to classification of eeg signals. The current work has introduced a preprocessing pipeline useful in preparing EEG signals for classification duties incorporating deep learning techniques. This pipeline aggregates numerous fundamental preprocessing steps, which are inter hierarchically constructed to enhance the data quality for the subsequent model training and evaluation.

Data Decoding:

The first function of the preprocessing pipeline is involved with decoding of EEG signal data, a core step that forms the basis of data conversion from raw binary format into comprehensible tensors.

File Analysis

The pipeline starts with the acquisition of data stored in .Npy files, thus, using TensorFlow's capabilities of analyzing files and accessing the encoded data efficiently.

Raw Data Decoding

As a result, the raw binary data pass through some process of decoding with the use of TensorFlow's `tf.io.decode_raw` function, which assists with the decoding of the binary stream into a float32 tensor in this case.

Tensor Restructuring

Following decoding, the tensor undergoes reshaping to match the input dimensions for compatibility with the next processing step and desired model structures.

Subsampling and Padding

In this case, subsampling operations based on provided offsets (if any) can be performed on the decoded tensor, which allows for extraction of labeled subsamples from the entire spectrogram. Additionally, the padding is applied while making sure that the signal dimensions are the same, thus ensuring data consistency.

Logarithmic Transformation

Another important processing in signal preprocessing that is carried out is the conversion of the spectrogram data through logarithmic transformation. Ahead, this transformation helps in negating the wide dynamic range which is observed in the EEG signals to ease the extraction of features and model convergence.

Normalization

After taking logarithm of the spectrogram data, some normalization procedures are applied to bring the data closer to normalized distribution. This normalization step is important to maintain stability throughout the training phase and to have the model able to differentiate between the important features required from the input data.

Channel Conversion

Following model requirements and to benefit from pre-trained weights of other models trained on RGB images, the one channel spectrogram is expanded into three channels. It also enables simple compatibility with deep learning structures designed to handle multiple channel inputs.

Data Augmentation

Regularization techniques are quite useful in adding variability and diversity to a training set to make models more reliable and capable of optimizing generalized performance. The augmentation stage encompasses the following methodologies:

Mix-up

Mix-up augmentation means that two samples are combined as pairs of samples that create some augmented training instances. By exposing the learner to data from varied distributions, as is the case with this technique, it becomes less prone to overfitting.

Random Cutout (Frequency-Masking)

Frequency-masking augmentation involves the random masking of certain frequency components in the spectrogram of the sentence. This mimicry of occlusions or noise helps to build up the model's robustness to variations in a signal stream and it also helps the model to identify important characteristics that should be extracted from the signal stream despite the presence of any distortions.

Random Cutout (Time-Masking)

The temporal information is included by performing random cutouts, in other words, temporal segments in the spectrogram are disguised. This augmentation strategy helps the stability of the model as it advances various temporal patterns and forces the model to learn the invariants within these patterns.

Dataset Construction

The dataset construction phase encompasses the following sub-stages: The dataset construction phase encompasses the following sub-stages:

Tensor Slicing

The process of creating the dataset means splitting different provided paths, offsets, and labels, which form tuples, each of which corresponds to the sample.

Mapping Decoding Function

Through decoding, each sample tuple decoctions through decoding, converting raw data into interpretable tensors that have been optimized for use in a number of other training tasks.

Caching

Intermediately, the decoded data may be stored to speed up another iteration; this reduces the burden of data loading during the cycle and improves the efficiency of the overall training process.

Repetition

For extended data coverage while training and making the dataset repeatedly expose multiple instances, this might be optionally set to a value of 0, which means indefinite.

Shuffling

Random samples are made from the data to ensure variation to avoid overfitting and to grant the model capacity to accurately predict various distribution of data.

Batching

This again enhances efficiency in model training through computational resources management during model training and gradient descent optimization and sample batching constitutes combining them in batch sizes that are predetermined, manageable, and feasible during the model training.

Augmentation Mapping

Use of data augmentation techniques where applicable brings variation and novelty into the training dataset hence strengthening the underlying model against certain forms of bias and improving the model's adaptability.

Prefetching

When batching is done, data preprocessing and model execution can be overlapped which makes training pipeline more efficient, reducing idle time and unnecessary usage of resources.

The preprocessing stages that have been demonstrated in this project result in a proper model that can be adopted for stage by stage preparation of EEG data for classification utilizing deep learning algorithms. This precise process guarantees trustworthy data and a more reliable and robust model making the task of analyzing EEG signals and classifying them easier to achieve in clinical or research environments.

Deep Neural Network Architecture

The architecture is composed of an ensemble of three deep neural network models - original EfficientNet B0, hyperparameter optimized 5-fold EfficientNet B1 and an EfficientNet B1 loaded with the spectrogram images generated during the preprocessing stage and an explainable model for understanding the

performance of the model in detail. The mean of all predictions from the three models of the ensemble is considered as the final prediction.

EfficientNet B0 Architecture

Layer	Details
Conv2d	Input: 1 channel, Output: 32 channels, Kernel Size: (3, 3), Stride: (2, 2), Padding: (1, 1), Bias: False
BatchNormAct2d	Channels: 32, Epsilon: 1e-05, Momentum: 0.1, Affine: True, Track Running Stats: True
Sequential	-
Sequential	-
Depthwise Separable Conv	-
Conv2d	Input: 32 channels, Output: 32 channels, Kernel Size: (3, 3), Stride: (1, 1), Padding: (1, 1), Groups: 32, Bias: False
SqueezeExcite	-
Conv2d	Input: 32 channels, Output: 8 channels, Kernel Size: (1, 1), Stride: (1, 1)
Conv2d	Input: 8 channels, Output: 32 channels, Kernel Size: (1, 1),

Layer	Details
	Stride: (1, 1)
Conv2d	Input: 32 channels, Output: 16 channels, Kernel Size: (1, 1), Stride: (1, 1), Bias: False
Sequential	-
Inverted Residual	-
Conv2d	Input: 16 channels, Output: 96 channels, Kernel Size: (1, 1), Stride: (1, 1), Bias: False
Conv2d	Input: 96 channels, Output: 96 channels, Kernel Size: (3, 3), Stride: (2, 2), Padding: (1, 1), Groups: 96, Bias: False
SqueezeExcite	-
Conv2d	Input: 96 channels, Output: 24 channels, Kernel Size: (1, 1), Stride: (1, 1), Bias: False
Sequential	-
Inverted Residual	-
Conv2d	Varies

Layer	Details
Conv2d	Varies
SqueezeExcite	-
Conv2d	Varies
Conv2d	Input: Varies, Output: 1280 channels, Kernel Size: (1, 1), Stride: (1, 1), Bias: False
BatchNormAct2d	Channels: 1280, Epsilon: 1e-05, Momentum: 0.1, Affine: True, Track Running Stats: True
SelectAdaptivePool2d	Pool Type: avg, Flatten: Flatten(start_dim = 1, end_dim=-1)
Linear	Input Features: 1280, Output Features: 6, Bias: True

EfficientNet B1 fold architecture

Layer	Details
Stem	Conv2d(1, 32, kernel_size=(3, 3), stride=(2, 2), padding=(1, 1), bias = False)
	BatchNormAct2d(32)
Block 1	DepthwiseSeparableConv(
	Conv2d(32, 32, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), groups = 32, bias = False)
	SqueezeExcite(
	Conv2d(32, 8, kernel_size=(1, 1), stride=(1, 1)), Conv2d(8, 32, kernel_size=(1, 1), stride=(1, 1)), Sigmoid()
	Conv2d(32, 16, kernel_size=(1, 1), stride=(1, 1), bias = False)
Block 2	InvertedResidual(
	Conv2d(16, 96, kernel_size=(1, 1), stride=(1, 1), bias = False)
	Conv2d(96, 96, kernel_size=(3, 3), stride=(2, 2), padding=(1, 1), groups = 96, bias = False)
	SqueezeExcite(
	Conv2d(96, 4, kernel_size=(1, 1), stride=(1, 1)), Conv2d(4, 96, kernel_size=(1, 1), stride=(1, 1)), Sigmoid()
	Conv2d(96, 24, kernel_size=(1, 1), stride=(1, 1), bias = False)
Block 3	InvertedResidual(
	Conv2d(24, 144, kernel_size=(1, 1), stride=(1, 1), bias = False)
	Conv2d(144, 144, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), groups = 144, bias = False)
	SqueezeExcite(
	Conv2d(144, 6, kernel_size=(1, 1), stride=(1, 1)), Conv2d(6, 144, kernel_size=(1, 1), stride=(1, 1)), Sigmoid()
	Conv2d(144, 24, kernel_size=(1, 1), stride=(1, 1), bias = False)
Block 4	InvertedResidual(
	Conv2d(24, 144, kernel_size=(1, 1), stride=(1, 1), bias = False)
	Conv2d(144, 144, kernel_size=(5, 5), stride=(2, 2), padding=(2, 2), groups = 144, bias = False)
	SqueezeExcite(

Layer	Details
	Conv2d(144, 6, kernel_size=(1, 1), stride=(1, 1)), Conv2d(6, 144, kernel_size=(1, 1), stride=(1, 1)), Sigmoid()
	Conv2d(144, 40, kernel_size=(1, 1), stride=(1, 1), bias = False)
Block 5	InvertedResidual(
	Conv2d(40, 240, kernel_size=(1, 1), stride=(1, 1), bias = False)
	Conv2d(240, 240, kernel_size=(5, 5), stride=(1, 1), padding=(2, 2), groups = 240, bias = False)
	SqueezeExcite(
	Conv2d(240, 10, kernel_size=(1, 1), stride=(1, 1)), Conv2d(10, 240, kernel_size=(1, 1), stride=(1, 1)), Sigmoid()
	Conv2d(240, 40, kernel_size=(1, 1), stride=(1, 1), bias = False)
Block 6	InvertedResidual(
	Conv2d(240, 240, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), groups = 240, bias = False)
	SqueezeExcite(
	Conv2d(240, 10, kernel_size=(1, 1), stride=(1, 1)), Conv2d(10, 240, kernel_size=(1, 1), stride=(1, 1)), Sigmoid()
	Conv2d(240, 80, kernel_size=(1, 1), stride=(1, 1), bias = False)
Head	Conv2d(80, 480, kernel_size=(1, 1), stride=(1, 1), bias = False)
	BatchNormAct2d(480)
	AdaptiveAvgPool2d(1)
	Flatten()
	Linear(in_features = 480, out_features = 1000, bias = True)
	Dropout(p = 0.2, inplace = False)
	Linear(in_features = 1000, out_features = 1000, bias = True)
	Dropout(p = 0.2, inplace = False)
	Linear(in_features = 1000, out_features = 1000, bias = True)
	Linear(in_features = 1000, out_features = 10, bias = True)

HMS HBAC Dataset

All imaging data used in this research are obtained and augmented from the Harmful Brain Activity Classification Dataset provided by Harvard Medical School under the license CC BY-NC 4.0. The dataset consists of samples belonging to seizure, generalized periodic discharges, lateralized periodic discharges, lateralized rhythmic delta activity, generalized rhythmic delta activity, or “other”. The classes are labeled with their initials for convenience i.e., SZ, GPD, LPD, LRDA and GRDA respectively. The proficient annotators examined EEG samples spanning 50 seconds and matched them with spectrograms that highlighted a 10-second interval per minute and had been centered simultaneously; the central 10 seconds were annotated. These samples had numerous overlaps and have been consolidated. Each spectrogram image is resized to 400 x 300 resolution and is augmented with the corresponding EEG capture as shown in Fig. 2. The data is partitioned into 5 folds for allocation to training, validation, and testing sets. It is ensured that there is no overlap between the splits to prevent any data leaking problems.

Results and discussions

Based on the inference obtained from the test data, Kullback Liebler divergence score is used for calculating the difference between the observed probability and predicted probability. The pipeline achieved 0.30 Kullback Liebler divergence score on the test data.

Explanatory heat maps based on ensemble model prediction

Spike and sharp-wave discharges are brief, high-amplitude spikes or waves that appear abruptly in the EEG recording. They typically indicate abnormal neuronal firing and are often observed in epileptic seizures. Figure 3 shows that the spectrogram is classified as a seizure because the signal strength is higher for the top 50 units in each capture (at four different timespans). This implies that during specific segments of the EEG recording, there is a notable increase in the intensity or magnitude of the brain's electrical activity. This heightened signal strength is observed across four different timespans, indicating that the elevated activity persists throughout the recording and is not isolated to a single instance the concentration of heightened activity within the top 50 units suggests a localised burst of electrical activity within specific time intervals, rather than a continuous pattern throughout the entire recording. The EEG signals exhibit sharp, pronounced spikes with significantly elevated signal strength. These spikes are indicative of intense neuronal firing and are commonly associated with seizure activity. Despite the presence of buried waves, which represent underlying brain activity, the dominant feature observed is the presence of abnormal spikes. This suggests that the seizure activity conceal and masks the underlying normal brain activity. We can conclude by the behaviour of the signal that it indicates the seizure activity .

Generalised periodic epileptiform discharges (GPEDs) are a periodic electroencephalogram (EEG) pattern with bisynchronous sharp wave complexes occurring in periodic intervals between 0.5 and 4 seconds ¹. This EEG pattern has been described critically ill patients, often with metabolic or postanoxic

encephalopathies, convulsive status epilepticus (SE), and progressive central nervous system diseases, including Creutzfeldt–Jacob disease (CJD) or subacute sclerosing panencephalitis (SSPE) but GPED pattern are believed to predict a grave outcome with a poor prognosis. These are abnormal EEG patterns characterised by repetitive, often rhythmic, discharges that involve widespread areas of the brain. They are typically associated with underlying brain dysfunction and can occur in various neurological conditions. In GPDs, one of the defining features is the presence of bisynchronous sharp wave complexes. These complexes are characterised by sharp, high-amplitude waves that occur synchronously across both hemispheres of the brain. Signal strength analysis allows for the identification of these bursts and helps differentiate them from background EEG activity. The concentration of heightened activity within specific time intervals, such as the top 25 units in each capture, indicates localised bursts of GPD activity. Figure 4 shows that the spectrogram is classified as a GPD because bisynchronous sharp wave complexes are observed, and the signal strength is higher for the top 25 units in each capture. The classification of the EEG pattern as a GPD signifies poor neurological activity, often associated with underlying brain dysfunction. GPDs are frequently observed in comatose patients, reflecting compromised neural function and mild uncontrolled neural signals.

The presence of GPDs suggests a state of altered brain function and may serve as an indicator of neurological deterioration. Identifying localised bursts of electrical activity, observing temporal dynamics, comparing with background activity, and correlating with clinical features. These aspects collectively contribute to the interpretation and understanding of GPDs in EEG recordings.

Lateralized Rhythmic Delta Activity (LRDA): LRDA refers to rhythmic delta waves (a type of slow brain wave) that are localised to one hemisphere of the brain. Like LPDs, LRDA suggests focal brain dysfunction and can be seen in various neurological disorders.

Interictal scalp EEG showed intermittent 2.5-Hz lateralized rhythmic delta activity (LRDA) in the right frontotemporal region, synchronous with the right hippocampal epileptiform discharges on intracranial EEG. The association between seizures and LRDA is frequency-dependent: only frequencies ≥ 1.5 Hz are associated with seizure. Nearly all patients with LRDA have focal injury on the side of the rhythmic activity. Figure 5 shows that the spectrogram is classified as LRDA because abnormal signals are observed at very early stages i.e., starting from 10–15 units. LRDA is often transformed into acute seizures as there is a growth in the recurrent intervals of these abnormally high spikes in most of the LRDA cases. It is a common case in critically ill patients. LRDA is characterised by rhythmic delta waves localised to one hemisphere of the brain. In this case, the abnormal signals associated with LRDA are observed very early in the EEG recording, starting from 10–15 units. The early onset of abnormal signals suggests that LRDA is present from the beginning of the recording, indicating a persistent pattern of abnormal brain activity localised to one hemisphere. This change is demonstrated by an increase in the duration of the intervals of anomalous high blips in the LRDA pattern. The progressive increase in the frequency and intensity of abnormal spikes may point to the deterioration of the neurological abnormality and the beginning of seizure manifestation. It is worth highlighting that LRDA is reported frequently among the ICU patients. This vulnerability may stem from the patient's neurological

preexisting conditions or traumatic insults that place them at risk for developing abnormal EEG patterns, including LRDA, during their critical illnesses. LRDA can be considered as an adverse phenomenon in cases with critical patients as its presence shows the possible decline in neurological status and may serve as the marker of the severity of the patient's state. These findings give information on the spatial-temporal characteristics of LRDA in the clinical EEG, enabling proper analyzing and interpreting the abnormal brain activity states in patients with this disorder.

Imagine a situation in which synchronization or desynchronization occurs for higher frequencies, in other words, the level of the rhythms does not rise abruptly, which specifies that there are no sharp variations in activity, which might refer to seizures, periodic discharges or other known variations. However, the EEG signals look quite continuous and identical regarding the fluctuations of both frequency and amplitude. Spectrogram does not show any recognizable pathologic brain activity, like seizures, periodic discharges, or rhythmic delta activity that linked to the neurological disorders or disease. The lack of these atypical patterns may mean that the recorded EEG signals could be considered as physiologic fluctuations and variations or even completely benign findings that do not relate to any pathologic processes. Based on the analysis in Fig. 6, the spectrogram is categorized in the "Other" group as there is no sharp peak in the frequency available and the spectrogram does not demonstrate any prohibited damaging brain movements.

Generalised Rhythmic Delta Activity: This term describes rhythmic delta activity that involves widespread areas of both hemispheres of the brain. It can be a nonspecific finding but may indicate underlying brain pathology or dysfunction. Figure 7 shows that the spectrogram is classified as GRDA because abnormal signals are observed at early stages like 15–20 units and the spikes are significantly higher in frequency than the commonly observed level in LRDA cases. These are caused due to metabolic disturbances and cerebral lesions due to physical damage or insufficient supply of blood or oxygen to the brain. In this case, abnormal signals associated with GRDA are observed early in the EEG recording, starting from 15–20 units. Frontally dominant generalised rhythmic delta activity, previously referred to as frontal intermittent rhythmic delta activity (FIRDA), is a 2 to 3 Hz high amplitude rhythmic to semi-rhythmic activity with anterior predominance. This pattern was commonly felt to be a diagnostic indicator of a midline cerebral pathology, like a 3-ventricle region tumour, but is now well recognized as a non-specific finding in encephalopathic EEGs. It is associated with encephalopathic states due to various etiologies like toxic, metabolic, infectious, neoplastic, and epileptic entities. GRDA, with a posterior emphasis that is commonly referred to as occipital intermittent rhythmic delta activity (OIRDA), is commonly seen in children with absence epilepsy. These abnormal signals appear as soon as the recording starts and therefore, even in this highly limited view of the data, GRDA is evident from the onset and appears to be an ongoing issue that impacts the right and left hemispheres of the brain. The statement points out that the deposits in GRDA have higher frequencies compared to Lateralized Rhythmic Delta Activity (LRDA) with much higher frequency. This leads to GRDA having abnormally higher delta waves frequency or a faster rhythm as compared to the normal delta waves present in the LRDA, thus showing more aggressive pattern of abnormal impulses in the brain. GRDA is believed in patients with metabolic disorders and in cases of various cerebral pathology originating from physical injuries, blood or oxygen

deficiency in the brain. Non-epileptic encephalopathies are known to disrupt normal functioning of the brain and result in pathological on EEG forms like GRDA due to metabolic problems like electrolyte imbalances, or metabolic encephalopathy. Structural brain abnormalities, due to Cerebral Lesions, travel, or other conditions such as traumatic Brain Injury or stroke, can also play an important role in the development of GRDA. further evidence for the classification of the spectrogram as GRDA includes the appearance of the abnormally low signals in the bilateral preterm at an early age, higher frequency spikes than those of the LRDA, and metabolic changes in the cerebral lesion. These observations give information on the pathophysiological characteristics and clinical relevancy of such abnormal activity as GRDA in EEG recordings in analyzing and comprehending pathological signal change relevant to specified neurological disorders.

Declarations

Author Contribution

A.M formulated idea T.S Collected data and validated with results C.N did literature review

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Figures

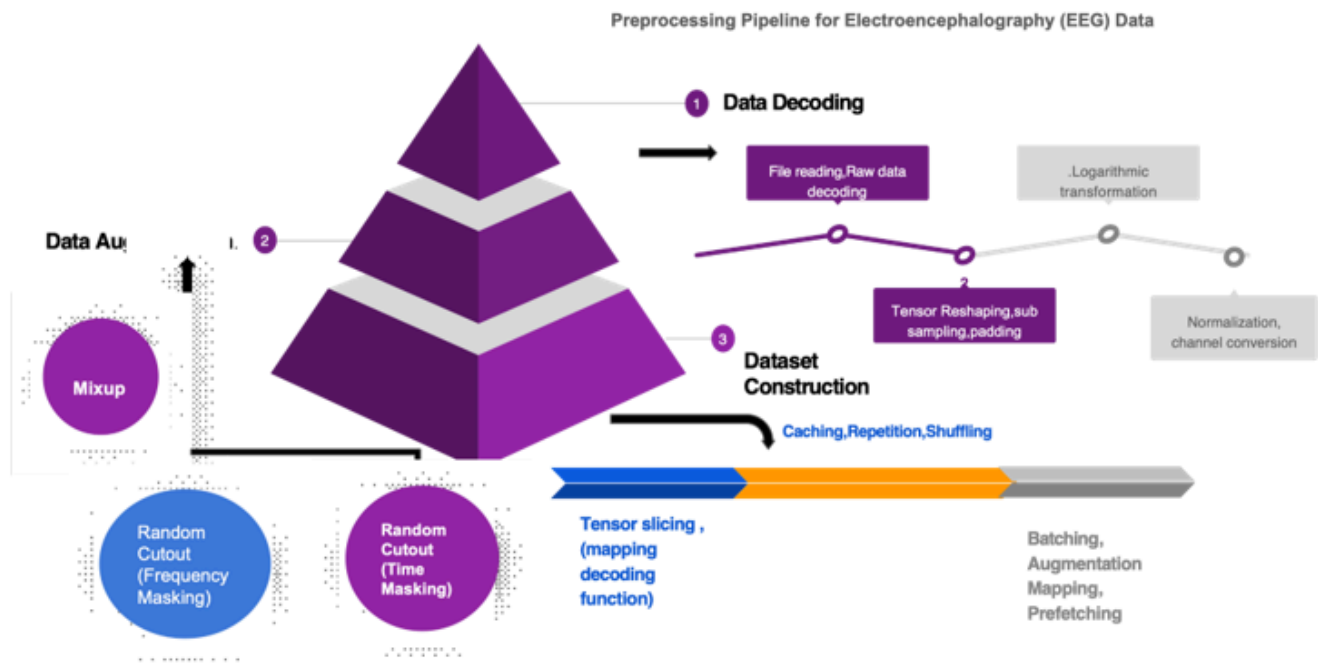


Figure 1

Preprocessing pipeline for electroencephalography (EEG) data

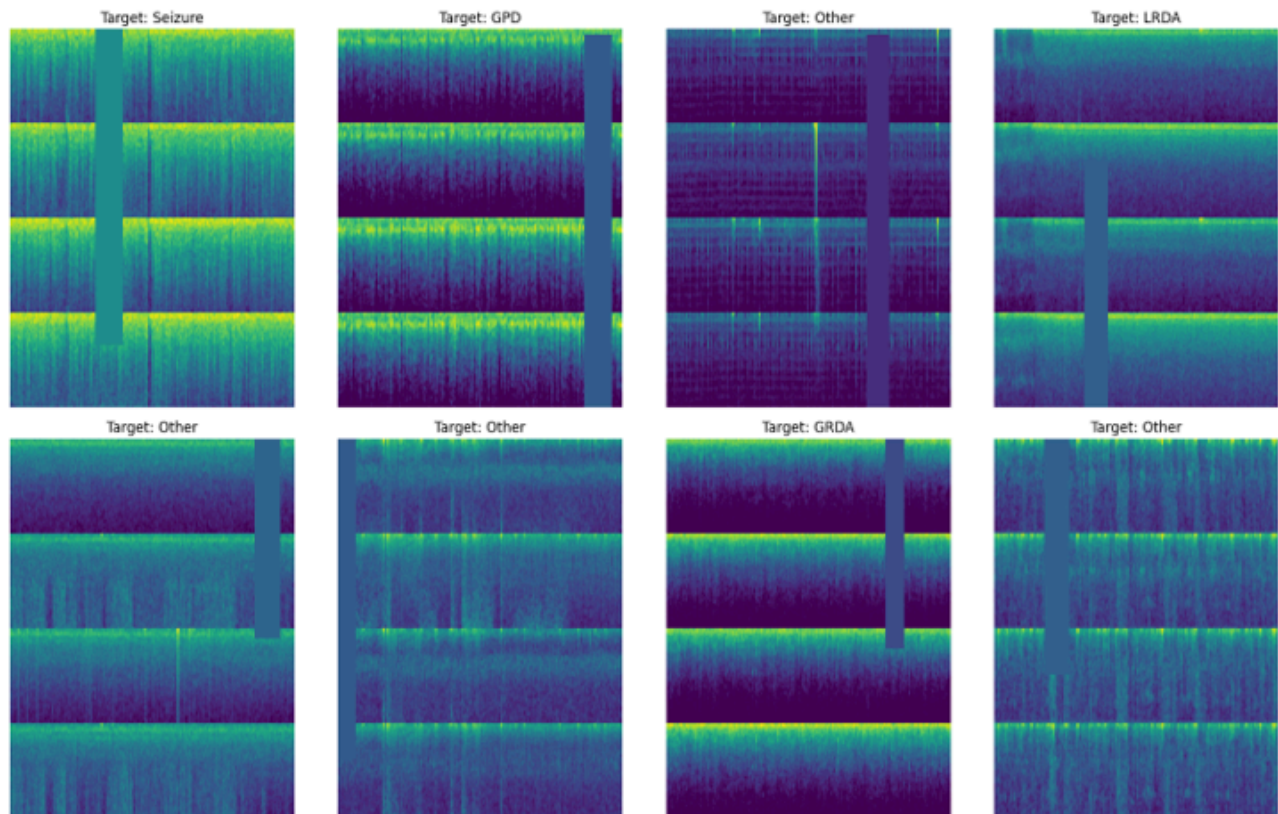


Figure 2

Sample spectrogram images belonging to different classes of harmful brain activity in the dataset.

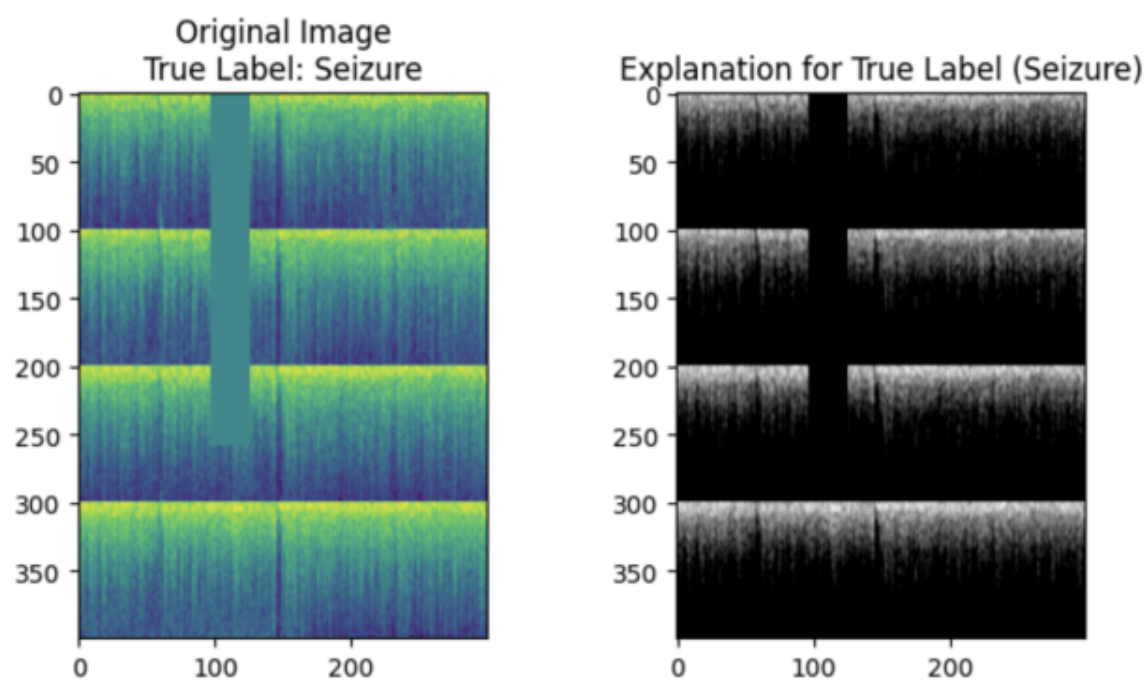


Figure 3

Recorded seizure signal and it's explainable heat map showing seizure patterns

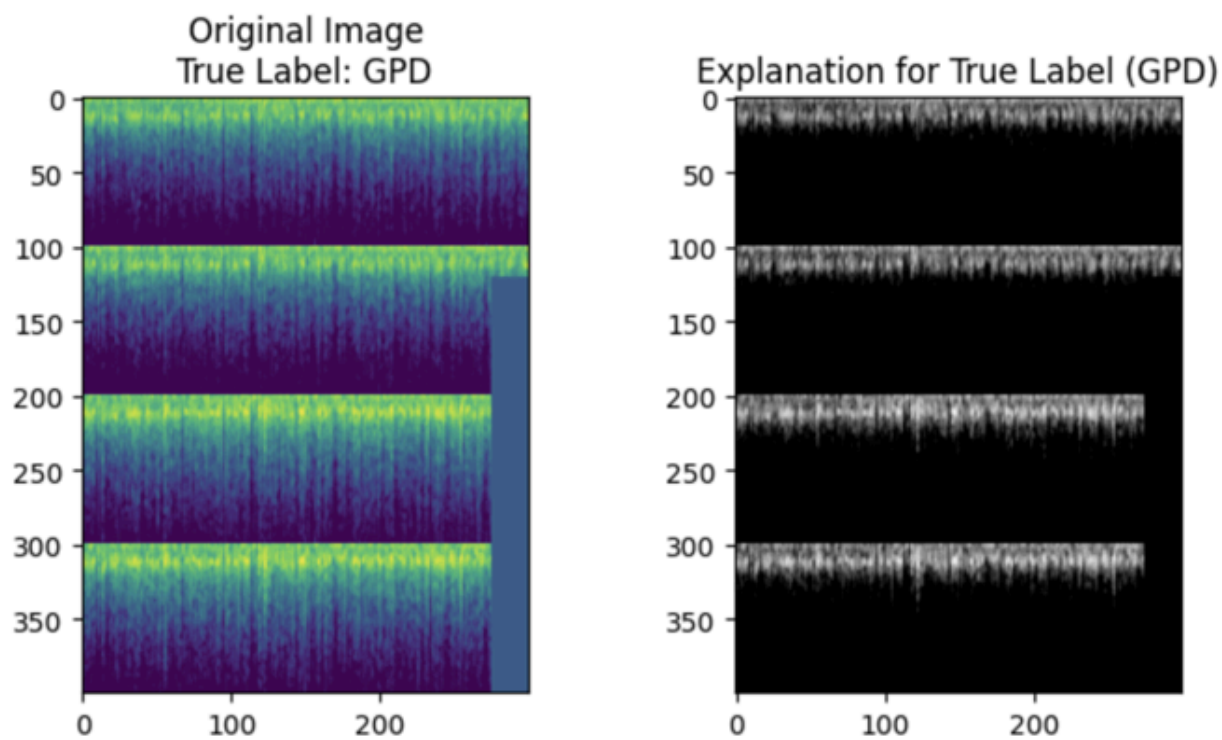


Figure 4

Recorded GPD signal and it's explainable heat map showing GPD patterns

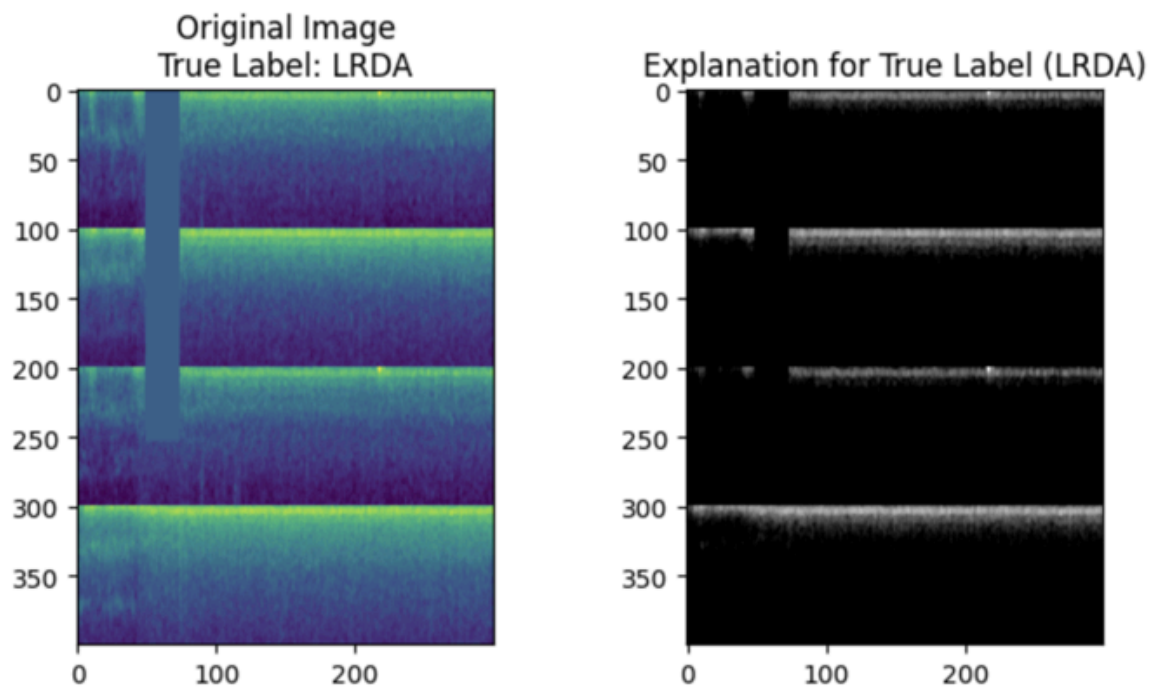


Figure 5

Recorded LRDA signal and it's explainable heat map showing LRDA patterns

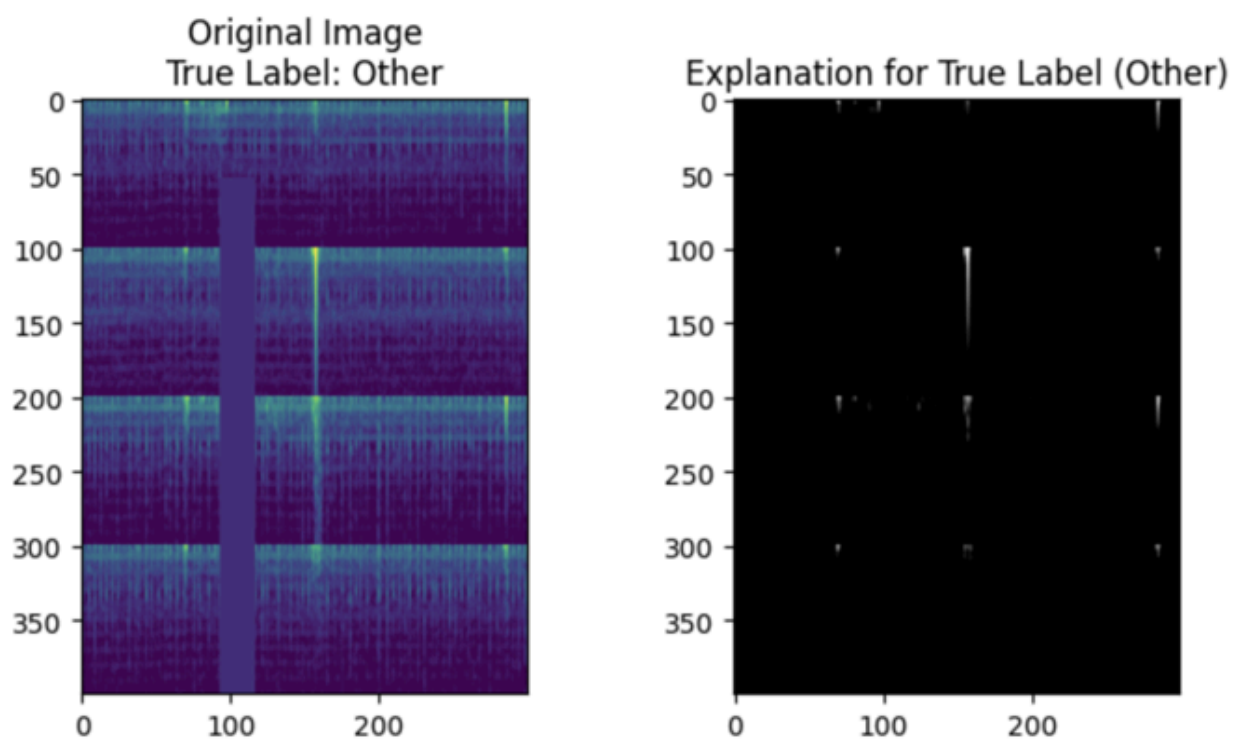


Figure 6

Recorded “other” signal and it’s explainable heat map showing no patterns pertaining to the HBAC classes

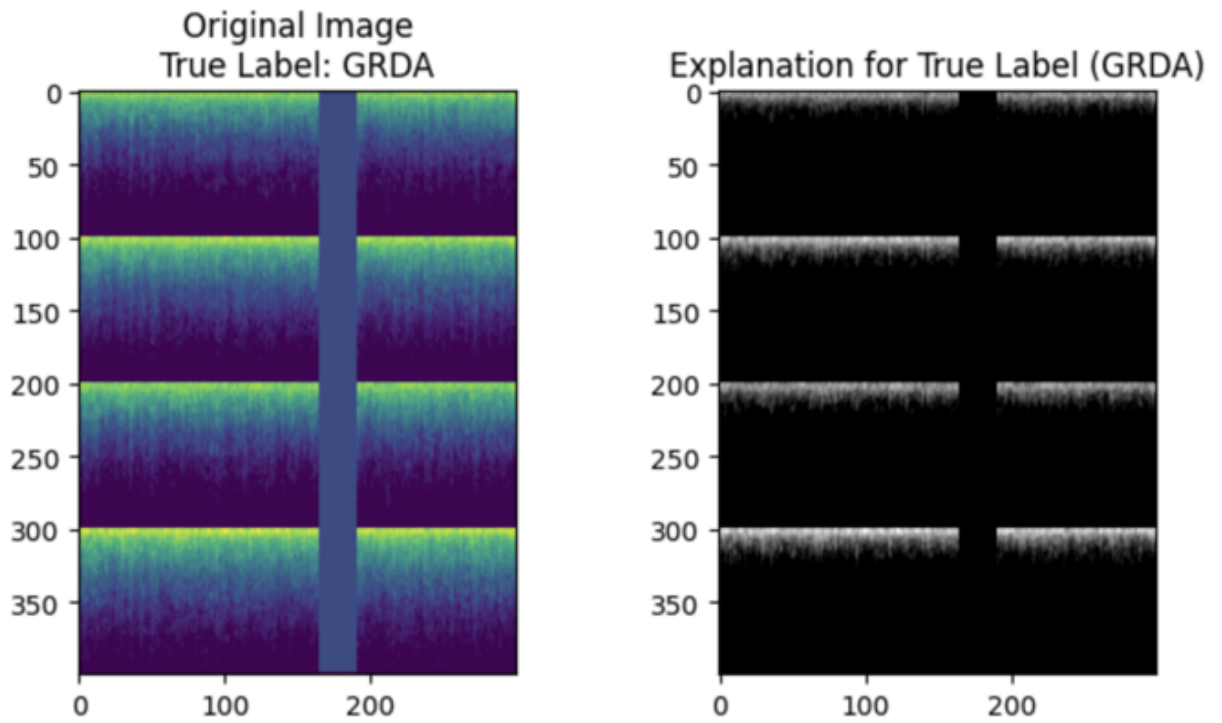


Figure 7

Recorded GRDA signal and it's explainable heat map showing GRDA patterns